

Workflow Documentation

Caleb Kennedy Hill

Summer 2026

Things to do:

- [Start pipeline docs](#)
- [Start metrics metadata](#)
- [Expand ArcGIS Portfolio docs](#)

This document is intended to serve as a reference for those of us working on the UNH Sustainability Institute's Split Incentive Tak Force.

1 Data Pipeline

This section will serve as documentation for the data pipeline. The required python libraries include `pandas`, `us`, and `census`. There are some others that are standard.

1.1 Setup

The preamble/defining involves setting the dictionaries `target_states` and `variables`, and the lists `target_municipalities` and `years`. The `us.states` module takes care of the FIPS numbers for each state. Target municipalities (i.e., NEMS members) are taken from a master list; formatting them correctly is a great use for a chatbot. This output needs to be verified, though. The years of interest will depend on the application; however, one should not use overlapping ACS5 datasets! [Click here to see why.](#)

1.2 Getting ACS5 Data

Currently there are [two different API calls](#) taking place: one for housing age-related data, and one for the rest. This will change. With modularity in mind, we should move to [separate calls for all different classes](#) of data. E.g., a call for population, a call for housing age, a call for heating fuel, etc.

1.3 Getting GIS Data

2 ArcGIS Online

The goal of this section is to detail the process of turning `.shp` files derived from `GeoDataFrame` into ArcGIS Instant Apps, for use in Portfolio projects. The system schematic is as follows:

Portfolio ← Instant App ← Web Map ← Feature Layer

IMPORTANT: To do any of this you will need an [ArcGIS Online account with publishing permissions](#). UNH has a license for ArcGIS Online, but we need to contact admin to get permissions. Let me know if you don't have publishing permissions.

Create a Web App

A **Portfolio** is a collection of **sections** which link to Web Apps. We will be tying into the efforts of and extending the work of BU's CELT program. With this in mind, our prototypical example of a Portfolio should be their Legacy Transition Atlas, which is linked [here](#).

1. Prepare ShapeFiles. Be sure that the relevant **GeoDataFrame** has been exported and compressed into a **.zip** file. The process of obtaining the shapefiles is outlined in Section 1.
2. Upload the **.zip** file to ArcGIS Online as a **Feature Layer**. Navigate to the **Content** tab of ArcGIS Online, then click **New item**. Click the **Feature layer**, and then the **Upload a file** options. Be sure to choose the upload option that creates a Hosted Feature Layer.¹ Pick a good name, and add a helpful description.
3. Back in the **Content** window, open the Hosted Feature Layer you just created. Choose the option to **Open in Map Viewer** in the upper right.
4. Fiddle and Save. The righthand side of the screen has a menu for changing how the map looks. We don't yet have a standardized way to approach this, so for now just do what you think is reasonable. When you're done, click **Save as** from the lefthand menu and pick the same name you used for the Feature Layer; add a short description as well.
5. In the **Content** window, open Web Map. In the options on the right, click **Create app** and choose **Instant App**. In the page that opens, choose the **Basic (Media Map)** option. Name it the same thing as the previous two.
6. In **Step 1** of the following screen, add the map you created in the previous step. Make sure a legend is made available in **Step 2**. So far, not much else has been standardized, so play around and pick something that doesn't seem crazy.
7. Before publishing, change the **Share** settings on the lefthand menu to public. Then Publish the app.

Create a Portfolio

A portfolio is the housing for several different Instant Apps.

Need to learn more before writing this up.

Portfolio ← Instant App ← Web Map ← Feature Layer

1. Create sections.
2. Upload base map for each section.
3. Put links to other apps in descriptopn.

¹This step is not possible without publishing permissions!